



SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)
Department of Mathematics and Computer Science

CIE-III SOLUTIONS

Topics: Joint Probability Distribution & Descriptive Statistics

UDSC-201: Probability and Statistics	Instructor: Dr. D Bhanu Prakash	Date: 06-10-2026
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1. The sample median is generally more resistant to outliers than the sample mean. [True/False | 1 Mark | BTL-1]

Answer: True

2. If it exists in a neighbourhood of $t = 0$, the moment-generating function of a random variable uniquely determines its probability distribution. [True/False | 1 Mark | BTL-2]

Answer: True

3. In a box plot, the box itself represents the _____ of the data, i.e. the middle 50% of observations. [1 Mark | BTL-2]

Answer: interquartile range (IQR)

4. If X and Y are independent random variables, then the moment-generating function of $X + Y$, written $M(X+Y)(t)$, is given by _____. [1 Mark | BTL-2]

Answer:

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$$

(The MGF of a sum of independent random variables equals the product of their individual MGFs.)

5. Match the graphical/numerical tool in Column A with its primary purpose in Column B. [2 Marks | BTL-2]

Column A	Column B
(a) Stem-and-leaf diagram	1. Assess the relationship between two quantitative variables
(b) Scatter diagram	2. Assess whether data follow a specified distribution (e.g. Normal)
(c) Probability plot	3. Displays the actual data values while showing the shape of the distribution
(d) Histogram	4. Displays the frequency of data using bars over class intervals

Answer: (a) $\rightarrow$ 3, (b) $\rightarrow$ 1, (c) $\rightarrow$ 2, (d) $\rightarrow$ 4

Q6. (a) Define the moment-generating function (MGF) of a random variable X and state its uniqueness property. (1) (b) Name any two commonly used joint probability distributions and give one real-life example of each. (1) [2 Marks | BTL-1]

Answer:

(a) The moment-generating function of X is defined as: $M_X(t) = E(e^{tx})$ provided the expectation exists for t in some open interval containing 0.

Uniqueness property: If two random variables have the same MGF (existing in a neighbourhood of 0), then they have the same probability distribution — the MGF uniquely characterises the distribution.

(b) Bivariate Normal distribution — e.g. the joint distribution of the height and weight of individuals in a population.

Multinomial distribution — e.g. the joint distribution of the counts of different coloured balls drawn in n draws from an urn containing several colours.

Note — General Functions of Random Variables

For a monotonic, differentiable function $Y = g(X)$ of a continuous random variable X , the pdf of Y is

obtained by the transformation rule: $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$

Alternatively, the expected value of any function $g(X)$ can be found directly — without first deriving the distribution of $Y = g(X)$ — using the Law of the Unconscious Statistician (LOTUS):

$$E[g(X)] = \int g(x) f_X(x) dx$$

Q7. The moment-generating function of a random variable X is given as

$M(t) = (1 - t)^{-1}$, $t < 1$. Using the properties of the MGF, find $E(X)$, $E(X^2)$ and $\text{Var}(X)$. [3

Marks | BTL-2]

Answer:

$$M(t) = (1 - t)^{-1}, \quad t < 1$$

Differentiating $M(t)$ with respect to t :

$$M'(t) = (1 - t)^{-2} \Rightarrow E(X) = M'(0) = 1$$

$$M''(t) = 2(1 - t)^{-3} \Rightarrow E(X^2) = M''(0) = 2$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 2 - 1 = 1$$

(This is recognised as the MGF of the Exponential(1) distribution, which is consistent with Mean = 1 and Variance = 1.)

Q3. (a) X and Y are independent random variables with $E(X) = 10$, $\text{Var}(X) = 4$, $E(Y) = 15$, $\text{Var}(Y) = 9$. If $W = 2X - 3Y$, find $E(W)$ and $\text{Var}(W)$. (2) (b) The marks of 10 students are: 5, 7, 7, 9, 10, 12, 13, 15, 18, 20. Compute the mean, median, range and interquartile range (IQR) of this data. (2)

[4 Marks | BTL-3]

Answer:

(a) For any random variables, the mean and variance of a linear combination are given by:

$$E(aX + bY) = aE(X) + bE(Y)$$

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

Since X and Y are independent, $\text{Cov}(X, Y) = 0$, so:

$$E(2X - 3Y) = 2E(X) - 3E(Y) = 2(10) - 3(15) = -25$$

$$\text{Var}(2X - 3Y) = 4 \text{Var}(X) + 9 \text{Var}(Y) = 4(4) + 9(9) = 16 + 81 = 97$$

(b) Arranged data: 5, 7, 7, 9, 10, 12, 13, 15, 18, 20 ($n = 10$)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{Mean} = (5+7+7+9+10+12+13+15+18+20) / 10 = 116 / 10 = 11.6$$

$$\text{Median} = \text{average of the 5th and 6th values} = (10 + 12) / 2 = 11$$

$$\text{Range} = \text{Maximum} - \text{Minimum} = 20 - 5 = 15$$

$$\text{Lower half } \{5, 7, 7, 9, 10\} \Rightarrow Q_1 = \text{median of lower half} = 7$$

$$\text{Upper half } \{12, 13, 15, 18, 20\} \Rightarrow Q_3 = \text{median of upper half} = 15$$

$$IQR = Q_3 - Q_1$$

$$IQR = Q_3 - Q_1 = 15 - 7 = 8$$

Q9. The joint pdf of X and Y is given as $f(x, y) = 2$, $0 < x < y < 1$ (a common joint distribution over a triangular region). (a) Show that $f(x, y)$ is a valid joint density. (1) (b) Find the marginal densities $f_X(x)$ and $f_Y(y)$. (1) (c) Find $\text{Cov}(X, Y)$. (1) (d) Using the linear-combination formula, find $E(3X + 2Y)$ and $\text{Var}(3X + 2Y)$. (2) [5 Marks | BTL-4]

Answer:

$$f(x, y) = 2, \quad 0 < x < y < 1$$

(a)

$$\int_0^1 \int_0^y 2 \, dx \, dy = \int_0^1 2y \, dy = 1$$

and $f(x, y) = 2 \geq 0$ on the triangular region $0 < x < y < 1$. Hence $f(x, y)$ is a valid joint pdf.

(b)

$$f_X(x) = \int_x^1 2 \, dy = 2(1 - x), \quad 0 < x < 1$$

$$f_Y(y) = \int_0^y 2 \, dx = 2y, \quad 0 < y < 1$$

(c)

$$E(X) = \int_0^1 x \cdot 2(1 - x) \, dx = \frac{1}{3}$$

$$E(Y) = \int_0^1 y \cdot 2y \, dy = \frac{2}{3}$$

$$E(X^2) = \frac{1}{6}, \quad \text{Var}(X) = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{18}$$

$$E(Y^2) = \frac{1}{2}, \quad \text{Var}(Y) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}$$

$$E(XY) = \int_0^1 \int_0^y xy \cdot 2 \, dx \, dy = \frac{1}{4}$$

$$\text{Cov}(X, Y) = \frac{1}{4} - \frac{1}{3} \cdot \frac{2}{3} = \frac{1}{36}$$

(d) Using $E(aX + bY) = aE(X) + bE(Y)$ and $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab \cdot \text{Cov}(X, Y)$:

$$E(3X + 2Y) = 3\left(\frac{1}{3}\right) + 2\left(\frac{2}{3}\right) = \frac{7}{3}$$

$$\text{Var}(3X + 2Y) = 9 \text{Var}(X) + 4 \text{Var}(Y) + 12 \text{Cov}(X, Y) = \frac{9}{18} + \frac{4}{18} + \frac{12}{36} = \frac{19}{18}$$